

Quiz 12*

♀ Age 20 st yes

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$P(\text{age} \leq 30 | \text{yes}) \quad P(\text{st} = \text{yes}) \quad P(\text{buy} = \text{yes})$$

$$\frac{2}{9} \quad \frac{6}{9} \quad \frac{9}{14}$$

$$P(y|\hat{x}) = \frac{2}{9} \times \frac{6}{9} \times \frac{9}{14} = 0.095$$